Topic: Different Types of Waves, The superposition Principle, Energy and Power traveling wave along string, Interference of waves, Standing Waves and Resonance

Different Types of Waves

Waves can be categorized into three main types:

1. **Mechanical Waves**:
    - These waves are most familiar as we encounter them constantly.
    - Common examples include water waves, sound waves, and seismic waves.
    - They have two essential characteristics:
        - They are governed by Newton's laws.
        - They can exist only within a material medium such as water, air, or rock.
2. **Electromagnetic Waves**:
    - These waves are less familiar but widely used in daily life.
    - Common examples include visible and ultraviolet light, radio and television waves, microwaves, x-rays, and radar waves.
    - Unlike mechanical waves, electromagnetic waves:
        - Require no material medium to propagate.
        - Can travel through a vacuum, such as light waves from stars reaching Earth.
    - All electromagnetic waves travel through a vacuum at the same speed:

$$c = 299,792,458 \text{ m/s}.$$

3. **Matter Waves**:
    - These waves are commonly used in modern technology but are less familiar.

- Associated with fundamental particles such as electrons, protons, and even atoms and molecules.
- Since these waves are linked to matter particles, they are termed *matter waves*.

Mechanical Waves

Mechanical waves require a medium to propagate and obey Newton's laws of motion. They can be classified into two main types:

1. Longitudinal Waves

- In longitudinal waves, the particles of the medium vibrate **parallel** to the direction of wave propagation.
- Common examples:
  - Sound waves in air.
- The wave equation for a longitudinal wave is:

  $$s(x,t) = s_m \cos(kx - \omega t)$$

  where:
  - $s(x,t)$ is the displacement of the medium,
  - s_m is the maximum displacement (amplitude),
  - k is the wave number,
  - ω is the angular frequency.

2. Transverse Waves

- In transverse waves, the particles of the medium vibrate **perpendicular** to the direction of wave propagation.
- These waves consist of **crests** (high points) and **troughs** (low points).
- Common examples:
  - Water waves.
  - Light waves (although electromagnetic, they exhibit transverse wave properties).
- The wave equation for a transverse wave is:

  $$y(x,t) = A \sin(kx - \omega t)$$

  where:
  - $y(x,t)$ is the displacement of the wave,
  - A is the amplitude,
  - k is the wave number,
  - ω is the angular frequency.

Wave Equation for a Transverse Wave

A transverse wave is a mechanical wave where the displacement of particles is **perpendicular** to the direction of wave propagation.

The general equation for a sinusoidal transverse wave is:

$$y(x,t) = y_m \sin(kx - \omega t + \phi) \tag{1}$$

where:

- $y(x,t)$ is the displacement at position x and time t,
- y_m is the amplitude (maximum displacement),
- k is the **wave number**, given by:

$$k = \frac{2\pi}{\lambda} \tag{2}$$

  where λ is the wavelength,
- ω is the **angular frequency**, given by:

$$\omega = 2\pi f \tag{3}$$

  where f is the frequency,
- ϕ is the phase constant.

Frequency and Time Period

- The **frequency** of the wave is:

$$f = \frac{\omega}{2\pi} \tag{4}$$

  where f is measured in Hertz (Hz).
- The **time period** of the wave is:

$$T = \frac{1}{f} = \frac{2\pi}{\omega} \tag{5}$$

  which represents the time required for one complete oscillation.

Wave Speed and Its Derivation

The wave speed v is the speed at which the wave propagates through the medium. It is given by:

$$v = \frac{\lambda}{T} \tag{6}$$

Substituting $T = \frac{2\pi}{\omega}$ and $\lambda = \frac{2\pi}{k}$, we get:

$$v = \frac{\frac{2\pi}{k}}{\frac{2\pi}{\omega}} \tag{7}$$

Simplifying,

$$v = \frac{\omega}{k} \tag{8}$$

Thus, the wave speed can also be written as:

$$v = f\lambda \tag{9}$$

This equation shows that the wave speed depends on the frequency and the wavelength.

Summary

For a transverse wave:

- The wave number is $k = \frac{2\pi}{\lambda}$.
- The frequency is $f = \frac{\omega}{2\pi}$.
- The time period is $T = \frac{1}{f}$.
- The wave speed is $v = \frac{\omega}{k} = f\lambda$.

Wave Speed on a Stretched String

The speed of a wave on a stretched string is determined by the properties of the string rather than the properties of the wave, such as frequency or amplitude.

The speed v of a wave on a string with tension T and linear density μ is given by:

$$v = \sqrt{\frac{T}{\mu}} \tag{10}$$

To express the linear density μ, we consider the mass of a small element of the string. The linear density is defined as the total mass m of the string divided by its total length l:

$$\mu = \frac{m}{l} \tag{11}$$

ENERGY AND POWER OF A WAVE TRAVELING ALONG A STRING

The average power, or the average rate at which energy is transmitted by a sinusoidal wave on a stretched string, is given by:

$$P_{\text{avg}} = \frac{1}{2}\mu v \omega^2 A^2 \tag{12}$$

Problem 1:

A string has a linear density $\mu = 525$ g/m $= 0.525$ kg/m and is under a tension $T = 45$ N. A sinusoidal wave with frequency $f = 120$ Hz and amplitude $y_m = 8.5$ mm $= 0.0085$ m is sent along the string. Determine the average rate at which the wave transports energy.

Solution

The angular frequency:
The angular frequency ω is given by:

$$\omega = 2\pi f \tag{13}$$

Substituting the given values:

$$\omega = (2\pi)(120) = 754 \text{ rad/s} \tag{14}$$

The wave speed :
The wave speed v on a stretched string is given by:

$$v = \sqrt{\frac{T}{\mu}} \tag{15}$$

Substituting the given values:

$$v = \sqrt{\frac{45}{0.525}} = 9.26 \text{ m/s} \tag{16}$$

The average power
The average power transported by the wave is given by:

$$P_{\text{avg}} = \frac{1}{2}\mu v \omega^2 y_m^2 \tag{17}$$

Substituting the known values:

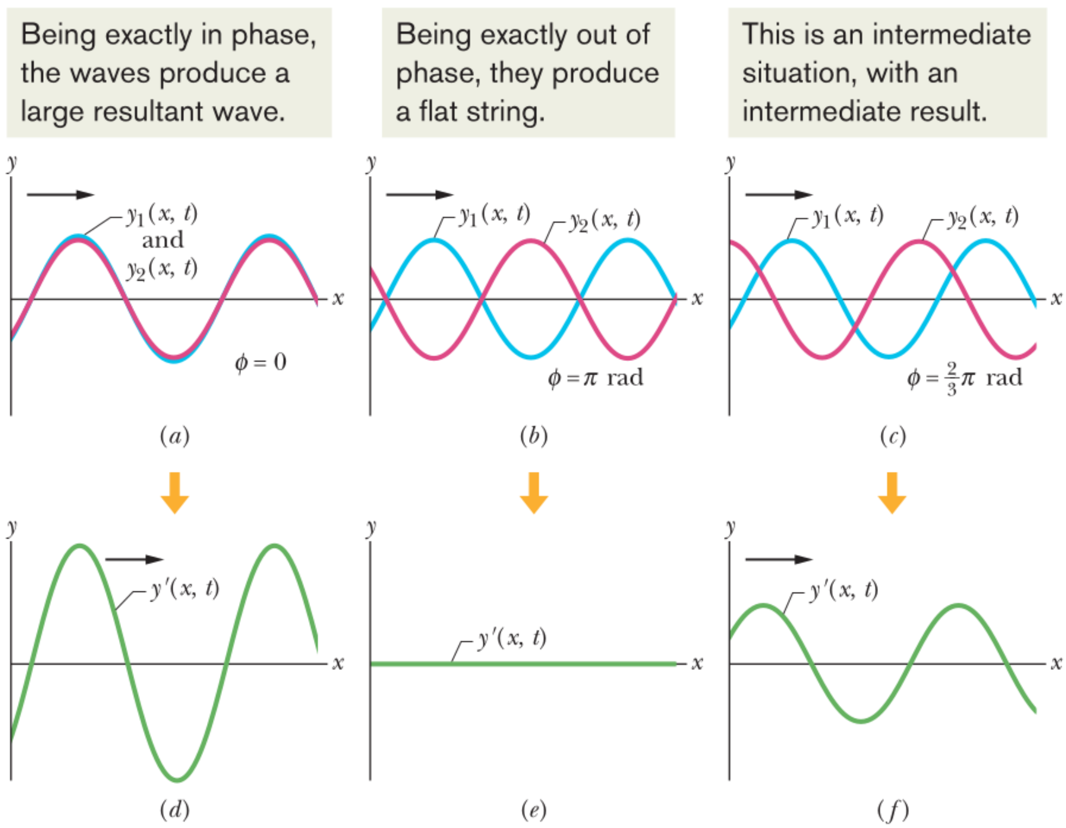
$$\begin{aligned} P_{\text{avg}} &= \frac{1}{2}(0.525)(9.26)(754)^2(0.0085)^2 \\ &= \frac{1}{2}(0.525)(9.26)(568516)(7.225 \times 10^{-5}) \\ &= \frac{1}{2}(0.525)(9.26)(41.08) \\ &= \frac{1}{2}(0.525)(380.5) \\ &= \frac{1}{2}(199.76) \\ &= 99.88 \approx 100 \text{ W} \end{aligned}$$

Final Answer: The average power transported by the wave is approximately 100 W.

Interference of Waves

Suppose we send two sinusoidal waves of the same wavelength and amplitude in the same direction along a stretched string. The superposition principle applies, and the resultant wave depends on the phase difference between them.

If the waves are exactly in phase (peaks and valleys aligned), they combine to double the displacement. If they are exactly out of phase (peaks of one align with valleys of the other), they cancel out, leaving the string straight. This phenomenon is called **interference**, and the waves are said to interfere.



Mathematical Representation

Let one wave traveling along a stretched string be given by:

$$y_1(x,t) = y_m \sin(kx - \omega t) \tag{18}$$

and another wave, phase-shifted by ϕ, be given by:

$$y_2(x,t) = y_m \sin(kx - \omega t + \phi). \tag{19}$$

The resultant wave, using the principle of superposition, is:

$$y'(x,t) = y_1(x,t) + y_2(x,t) \tag{20}$$

$$= y_m \sin(kx - \omega t) + y_m \sin(kx - \omega t + \phi). \tag{21}$$

Using the trigonometric identity:

$$\sin A + \sin B = 2\cos\left(\frac{A-B}{2}\right)\sin\left(\frac{A+B}{2}\right), \tag{22}$$

we obtain:

$$y'(x,t) = \left[2y_m \cos\left(\frac{\phi}{2}\right)\right] \sin(kx - \omega t + \frac{\phi}{2}). \tag{23}$$

Thus, the resultant wave remains sinusoidal and travels in the same direction with an amplitude:

$$y'_m = \left|2y_m \cos\left(\frac{\phi}{2}\right)\right|. \tag{24}$$

Special Cases

- If $\phi = 0$, the waves are exactly in phase, and we obtain:

  $$y'(x,t) = 2y_m \sin(kx - \omega t), \tag{25}$$

  meaning the amplitude is doubled.

- If $\phi = \pi$, the waves are exactly out of phase, resulting in complete destructive interference:

  $$y'_m = 0. \tag{26}$$

Problem 2:

Two identical sinusoidal waves, moving in the same direction along a stretched string, interfere with each other. The amplitude y_m of each wave is 9.8 mm, and the phase difference ϕ between them is $100°$.

(a) What is the amplitude y'_m of the resultant wave due to the interference, and what is the type of this interference?

(b) What phase difference, in radians and wavelengths, will give the resultant wave an amplitude of 4.9 mm?

Solution

(a) Amplitude of the resultant wave
The amplitude of the resultant wave is given by:

$$y'_m = \left| 2y_m \cos\left(\frac{\phi}{2}\right) \right| \tag{27}$$

Substituting the given values:

$$y'_m = \left| (2)(9.8 \text{ mm}) \cos\left(\frac{100^\circ}{2}\right) \right| \tag{28}$$

$$= |(19.6 \text{ mm}) \cos(50^\circ)| \tag{29}$$

Using a calculator:

$$y'_m = (19.6 \text{ mm})(0.6428) = 13 \text{ mm} \tag{30}$$

Type of interference: Since the phase difference is between 0° and 180°, the interference is intermediate.

(b) Finding the phase difference for a given resultant amplitude
Given that $y'_m = 4.9$ mm. Now:

$$4.9 \text{ mm} = \left| 2(9.8 \text{ mm}) \cos\left(\frac{\phi}{2}\right) \right| \tag{31}$$

Solving for ϕ:

$$\cos\left(\frac{\phi}{2}\right) = \frac{4.9 \text{ mm}}{(2)(9.8 \text{ mm})} \tag{32}$$

$$= \frac{4.9}{19.6} = 0.25 \tag{33}$$

Taking the inverse cosine:

$$\frac{\phi}{2} = \cos^{-1}(0.25) \tag{34}$$

Using a calculator:

$$\frac{\phi}{2} = 1.3181 \text{ rad} \tag{35}$$

$$\phi = 2(1.3181) = 2.636 \text{ rad} \approx 2.6 \text{ rad} \tag{36}$$

Since phase difference can be positive or negative:

$$\phi \approx \pm 2.6 \text{ rad} \tag{37}$$

Phase difference in wavelengths:

$$\frac{\phi}{2\pi} = \frac{2.636 \text{ rad}}{2\pi \text{ rad/wavelength}} \tag{38}$$

$$\approx \frac{2.636}{6.283} = 0.42 \text{ wavelengths} \tag{39}$$

Thus, the phase difference corresponds to approximately ± 0.42 wavelengths.

Standing Waves

When two sinusoidal waves of the same amplitude and wavelength travel in opposite directions along a stretched string, their interference produces a standing wave.

Mathematical Representation

The two traveling waves can be represented as:

$$y_1(x,t) = y_m \sin(kx - \omega t) \tag{40}$$

$$y_2(x,t) = y_m \sin(kx + \omega t) \tag{41}$$

Using the principle of superposition, the resultant wave is:

$$y'(x,t) = y_1(x,t) + y_2(x,t) = y_m \sin(kx - \omega t) + y_m \sin(kx + \omega t) \tag{42}$$

Using the trigonometric identity:

$$\sin A + \sin B = 2 \sin\left(\frac{A+B}{2}\right) \cos\left(\frac{A-B}{2}\right) \tag{43}$$

we obtain:

$$y'(x,t) = [2y_m \sin(kx)] \cos(\omega t) \tag{44}$$

This equation represents a standing wave because it does not describe a traveling motion. Instead, it indicates that each point on the string oscillates with an amplitude $2y_m \sin(kx)$.

Nodes and Antinodes

The amplitude is zero when:

$$\sin(kx) = 0 \tag{45}$$

which occurs at positions:

$$x = n\frac{\lambda}{2}, \quad n = 0, 1, 2, \ldots \quad \text{(Nodes)} \tag{46}$$

The amplitude is maximum when:

$$|\sin(kx)| = 1 \tag{47}$$

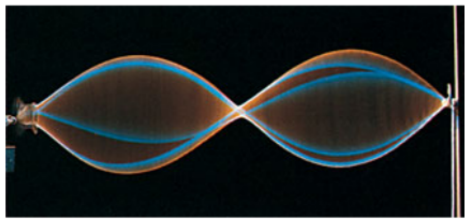
which occurs at positions:

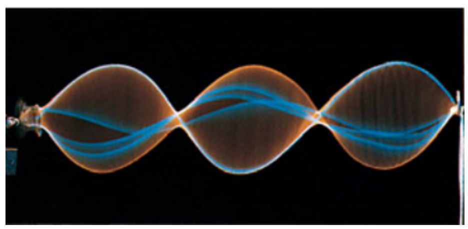
$$x = \left(n + \frac{1}{2}\right)\frac{\lambda}{2}, \quad n = 0, 1, 2, \ldots \quad \text{(Antinodes)} \tag{48}$$

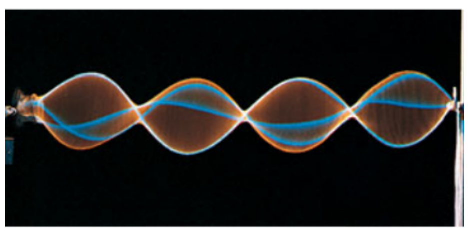
Nodes are points where the medium remains stationary, while antinodes are points of maximum displacement. Adjacent nodes and antinodes are separated by $\lambda/2$.

Resonance and Resonant Frequencies

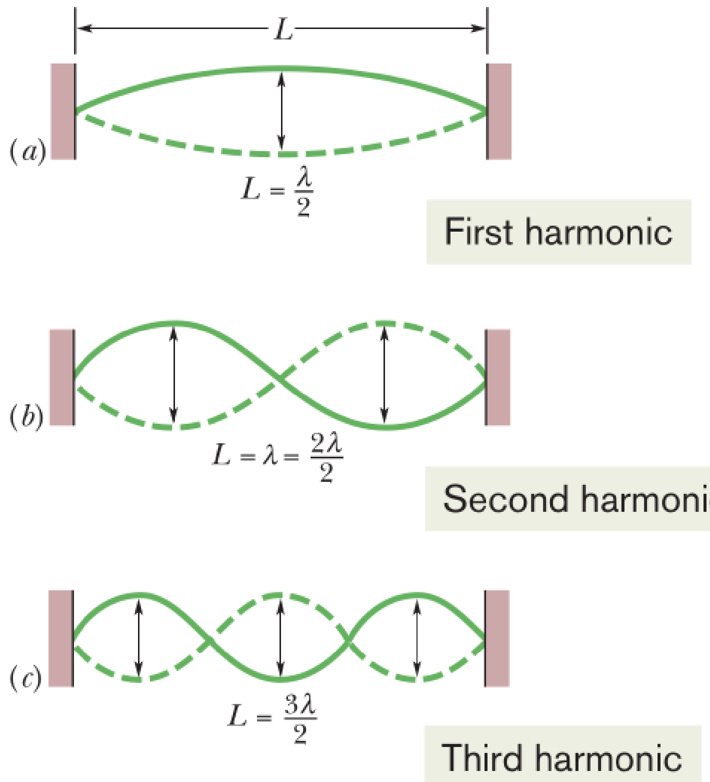
For certain frequencies, the interference produces a standing wave pattern (or oscillation mode) with nodes and large antinodes. Such a standing wave is said to be produced at resonance, and the string is said to resonate at these certain frequencies, called resonant frequencies.







Resonant Wavelengths and Modes



For a string of fixed length L with nodes at both ends, the standing wave condition requires that the wavelength satisfies:

$$\lambda_n = \frac{2L}{n}, \quad n = 1, 2, 3, \ldots \tag{49}$$

where n is the mode number. The fundamental mode ($n = 1$) has a wavelength $\lambda_1 = 2L$, the second harmonic ($n = 2$) has $\lambda_2 = L$, and so on.

Since the wave speed v is related to frequency and wavelength by $v = f\lambda$, the resonant frequencies are:

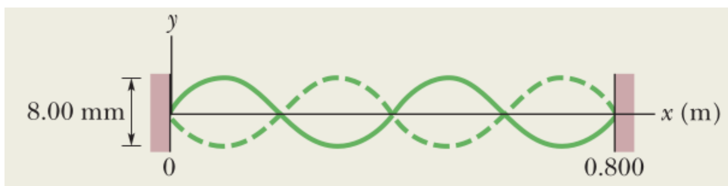
$$f_n = \frac{nv}{2L}, \quad n = 1, 2, 3, \ldots \tag{50}$$

Each successive mode has one additional node and antinode, fitting an additional half-wavelength into the length L.

Problem 3:

A string of mass $m = 2.500$ g and length $L = 0.800$ m is under tension $T = 325.0$ N. We need to determine:

1. **The wavelength λ of the transverse waves producing the standing wave pattern.**
2. **The harmonic number n.**
3. **The frequency f of the transverse waves and the oscillations of the moving string elements.**



Solution

Wavelength

From the given information, we see that 2 full wavelengths fit into the length L of the string:

$$2\lambda = L. \tag{51}$$

Solving for λ:

$$\lambda = \frac{L}{2} = \frac{0.800 \text{ m}}{2} = 0.400 \text{ m}. \tag{52}$$

Harmonic Number

By counting the number of loops (or half-wavelengths), we determine the harmonic number:

$$n = 4. \tag{53}$$

Wave Speed

The speed v of the transverse waves is given by:

$$v = \sqrt{\frac{T}{\mu}}, \tag{54}$$

where μ is the linear mass density:

$$\mu = \frac{m}{L} = \frac{2.500 \times 10^{-3} \text{ kg}}{0.800 \text{ m}} = 3.125 \times 10^{-3} \text{ kg/m}. \tag{55}$$

Substituting values:

$$v = \sqrt{\frac{325.0 \text{ N}}{3.125 \times 10^{-3} \text{ kg/m}}} = \sqrt{104000} \approx 322.49 \text{ m/s}. \tag{56}$$

Frequency Calculation

Using the wave equation:

$$f = \frac{v}{\lambda}, \tag{57}$$

we substitute:

$$f = \frac{322.49 \text{ m/s}}{0.400 \text{ m}} = 806.2 \text{ Hz} \approx 806 \text{ Hz}. \tag{58}$$

Alternatively, using the resonance condition:

$$f_n = \frac{nv}{2L}, \tag{59}$$

substituting values:

$$f_4 = \frac{4 \times 322.49}{2 \times 0.800} = 806.2 \text{ Hz} \approx 806 \text{ Hz}. \tag{60}$$